

Rehan Saqib

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PROFESSIONAL SUMMARY

Backend and DevOps-oriented Computer Engineering undergraduate with hands-on experience in containerization, Kubernetes orchestration, and AWS cloud deployments. Skilled in building RESTful APIs, implementing CI/CD pipelines, and deploying scalable cloud-native applications in production environments.

EDUCATION

National University of Science and Technology (NUST)

Bachelor of Engineering in Computer Engineering

Islamabad, Pakistan

Sept. 2024 – Sept. 2028

PROJECTS

Seedha Rasta – Fuel-Optimized Routing Engine | *Django, React, PostgreSQL, Redis, Docker, AWS ECS*

- Engineered a graph-based routing system using A* with a custom fuel cost function, reducing estimated fuel consumption by **18–25%** over shortest-path routing.
- Modeled traffic-aware routing using congestion and idle-time factors, improving real-world route accuracy and decision quality.
- Containerized services using **Docker** and deployed on **AWS ECS** with **Application Load Balancer** for scalable and fault-tolerant traffic handling.
- Built a **CI/CD pipeline with GitHub Actions** to automate build and deployment to AWS, reducing manual deployment effort by **70%**.
- Integrated **Amazon RDS (PostgreSQL)** and **AWS ElastiCache (Redis)**, reducing route computation latency by **40%**.

Synthron – AI-Powered PC Builder Platform | *Django, PostgreSQL, Docker*

- Built a full-stack AI-powered platform generating optimized PC builds based on user budgets, use cases, and regional constraints, improving recommendation relevance by **30%**.
- Designed RESTful APIs using **Django REST Framework** with secure JWT-based authentication, handling structured user queries and dynamic build generation.
- Implemented an AI-driven parsing pipeline (LLM-based) to extract user intent, reducing manual input requirements by **50%**.
- Containerized backend, frontend, and database using **Docker Compose**, reducing environment setup time by **70%**.
- Architected a modular multi-component system (CPU, GPU, RAM agents), improving recommendation efficiency and scalability by **25%**.

Cloud-Native Web Voting Application (DevOps Focus) | *Go, MongoDB, Docker, Kubernetes, AWS EKS*

- Deployed and managed a microservices-based application on **AWS EKS** to gain hands-on experience with Kubernetes orchestration.
- Worked with core Kubernetes primitives including **Namespaces, Deployments, Services, StatefulSets, PersistentVolumes, and Secrets**.
- Configured a **MongoDB replica set** within Kubernetes for high availability and fault tolerance.
- Focused on operational workflows such as debugging pods, managing cluster resources, and ensuring service reliability.

Todo List Application | *Django, PostgreSQL, Docker, AWS*

- Developed a backend-driven task management system with full CRUD functionality using Django.
- Containerized application and database services using **Docker** for portability and environment consistency.
- Deployed on **AWS EC2** and configured a **CI/CD pipeline using AWS CodePipeline, CodeBuild, and CodeDeploy**.
- Automated build and deployment workflows, enabling consistent and repeatable releases.

TECHNICAL SKILLS

Languages: Python, C++, Java

Backend: Django REST Framework

DevOps & Cloud: Docker, Kubernetes, AWS, GitHub Actions

Systems: Linux, Bash

Version Control: Git, GitHub